

Where does conda clean remove packages from?

Asked 4 years, 11 months ago Modified 3 months ago Viewed 61k times

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The conda documentation specifies:

conda clean [-h] [-y] [--dry-run] [--json] [--debug] [--verbose] [-q] [-a] [-i] [-t] [-p] [-s]

Remove unused packages and caches.

Example:

conda clean --tarballs

[\(conda clean documentation\)](#)

So it is suppose to remove *unused* packages.

where from ?

- from conda (base) ?
- from the active environment ?

Does the environment need to be active?

How does it know they are *unused* packages?

conda [Edit tags](#)

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edited Jan 31, 2019 at 13:36

asked Aug 22, 2018 at 5:50

 **jalazbe**
1,791 2 17 39

⬆️ There are some questions that need answer. Like how to know if they are unused? – [jalazbe](#) Jan 20, 2020 at 21:54

3 Answers

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This page and this one [Will "conda clean" erase my favorite packages?](#) give useful answers to ask before running this command, that's nice.

I learnt about hard/soft links, and one of us claims he actually dared cleaning his... stuff, that's a most precious info. I didn't dare push the button yesterday.

I would extend with the question around: why erasing packages, though they *might* look required (if you don't look at the versions around!), "**How does it know they are unused packages?**"

Well, that relates to why everybody uses conda : **because it is DAG based!**

If you didn't take the course upon that yet a quick overview here:

A DAG is a Directed Acyclic Graph, meaning a graph of modules/packages in conda. Directed means package A ---depends on----> B (Directed for only one direction here, that makes sense, and acyclic meaning we ensure we never have a cycle like in A->B, B->C, C->A). Ok, that's very obvious that it wouldn't work here thinking of what "depends on" means, at least in sytems, not everywhere (the same in your IDE if you code 3 methods calling another this way : you'll get an error from the compiler in Java, but this check is cheap, since you just have "a few" methods).

Well for managing this hard task of keeping your dependencies ok on conda, this "computational" graph is maintained all time, so the tool spends his life working on it, leveraging the insanely nice graph algorithms that just rule the world today.

When you push the clean cmd, you fire an algorithm that will traverse the whole graph (which also necessarily features your envs, pointing towards requested module ENV---requires--PACK) in an efficient way, keeping a count for each node, visited at least once (directly from the root), or more. At the end the guys with "1" are deleted.

The same is used for many applications : GC Garbage Collector in Java for instance is an intuitive similar problem : in our mind all the variables are useful, but actually not any more those that have no more context related to them (basically after for instance the completion of a method: returned values will keep living, but what about the helping auxiliary values created in your method?). The GB could not deal with them just after the method returns... why? Because it would be too computationally inefficient. A scheduled task is preferred, run upon under certain time or capacity constraints, to clean.

Well for conda, hearing about these "links" and knowing about the DAG technology inside, the parallel should now be clearer..

So what is deleted by conda is a **module/package in the DAG that is left all alone**, by itself (still seen, but only by the root of the graph of course because it used to be there and it needs to be cleaned one day, obviously).

Meaning this package was downloaded to be used one day, but a change happened on another (upgrade or downgrade for this dependency managing task), and another version of the package landed there around, and maybe all the packages everywhere ended up using the new vesion as well, until the day where the soon gone version was left with the root, alone.

But conda just like a GC in the JVM doesn't want to handle that when it happens : otherwise that would be another full check-run of the graph algorithm on your machine, each time just one dependency changes...

And so, the conda team left us our users run this task between two working days or two companies or when your drive gets full. Might make more sense, now that you know **you are like a GC yourself** (as me) in the loop :)

And also, the dry-run option lets you try it without executing it, to bring even more confidence in this leap of faith as in the doc:

[conda clean --all --dry-run](#)

Concepts related to Directed graphs (where checking for cycles is always a TODO thing for the alg to just work.

<https://github.com/kevin-wayne/algs4>

Credits : Princeton Bob Sedgewick's perfect video course from Princeton,

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edited Apr 9 at 5:33

 **Rich Lysakowski PhD**
2,675 31 43

answered Dec 11, 2022 at 19:25

 **RodKLV**
21 2

⬆️ Thanks for the reminding me that the dependency tree is a DAG and cannot (or "should not" be a cyclic graph. – [Rich Lysakowski PhD](#) Apr 9 at 5:37

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This is a very important question that deserves more discussion.

conda clean --all

will remove unused packages and caches.

It will delete all unused packages from ALL environments, not just the currently activated one.

This potentially frees up tens of GB of space. I have about 25 conda environments for different data science development and training projects. The Anaconda installation with all my conda environments ballooned to over 60GB of space.

Doing a `Remove` operation from within **Anaconda Navigator** did not actually free up the disk space.

Using `conda clean --all` removed unused packages, temp files, wheels, exe files and other binaries that were no longer used.

A Windows reboot was required to actually remove the unused packages.

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edited Apr 30, 2021 at 23:06

 **jalazbe**
1,791 2 17 39

answered Apr 29, 2021 at 5:15

 **Rich Lysakowski PhD**
2,675 31 43

2 ⬆️ Very interesting to know that it can use so much space. – [jalazbe](#) Apr 29, 2021 at 6:18

1 ⬆️ "It will delete all unused packages from ALL environments" – I'm not sure how this statement makes sense? If they packages are unused, doesn't that mean that they aren't installed in any environment? I guess you just mean it removes packages sitting around in caches that are unused by any/all environments? – [Ben Farmer](#) Oct 6, 2022 at 1:36 ✓

⬆️ I was having the same problem but a 'conda clean --all' fixed it for me. Why, I have no idea – [Pythonista anonymous](#) Apr 1 at 22:07

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It removes unused packages from under the `pkgs/` directory wherever you happen to have conda installed. An "unused" package is one that's not used in any environment. All conda packages are stored under the `pkgs/` directory and then hard-linked (if possible) into the environments.

As an aside, `conda clean` will print out the location of where the packages are actually located:

```
$ conda clean -all
Cache location: /data/processing/ryan/miniconda/pkgs
Will remove the following tarballs:
/data/processing/ryan/miniconda/pkgs
.....
filelock-3.0.10-py_0.tar.bz2                9 KB
```

Edit 13.3.2020 [py](#) pointed out that the `-all` option has been changed to `-a` or `--all` in conda 4.8.2.

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edited Mar 13, 2020 at 15:37

answered Feb 1, 2019 at 11:07

 **Devon Ryan**
819 9 9

6 ⬆️ Argument must be `-a` or `--all` as of conda 4.8.2 – [nrf](#) Mar 13, 2020 at 15:35

1 ⬆️ Thanks @nrf, I'll update the answer to include that. – [Devon Ryan](#) Mar 13, 2020 at 15:36

⬆️ @DevonRyan Does root/the base environment count as an environment for the purposes of determining whether a package is used in one? – [weirdalsuperfan](#) Aug 28, 2020 at 3:27

⬆️ @weirdalsuperfan yes – [Devon Ryan](#) Aug 28, 2020 at 5:52

⬆️ @DevonRyan You say "An "unused" package is one that's not used in any environment." But I don't understand what that means... are you able to explain in more detail? A `conda update --all` removes packages that are no longer required as dependencies for any other package, so is that something different from "unused"? – [cgb119](#) Jan 14, 2022 at 13:40 ✓

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https://stackoverflow.com/questions/5196539/where-does-conda-clean-remove-packages-from

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